

Garima Swami

AI Engineer · Generative AI & RAG Systems · Berlin, Germany

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PROFESSIONAL SUMMARY

AI Engineer and Artificial Intelligence Master's student at Brandenburg University of Technology, currently building production Generative AI systems at **Abacus Medicine**. Specializing in **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **agentic AI workflows**, and **workflow automation with n8n**. Hands-on with Python, LangChain, LlamaIndex, FastAPI, vector databases, and prompt engineering — with a strong software engineering foundation in Git, testing, Docker, and Agile delivery. Focused on shipping reliable AI systems that hold up in production, not just in demos.

WORK EXPERIENCE

Gen AI Engineer · Abacus Medicine

Aug 2026 – Present · Berlin, Germany

- Designing and deploying **agentic AI workflows** across pharmaceutical data operations, orchestrating multi-step automations with **n8n** for data ingestion, validation, and reporting.
- Building **LLM-powered RAG pipelines with LangChain**, grounding responses in structured pharmaceutical and supply-chain data sources for context-aware retrieval.
- Architecting production-grade generative AI systems with an emphasis on reliability, observability, and compliance in a regulated healthcare/pharma environment.
- Developing agent orchestration logic for multi-step task automation, tool-calling, and human-in-the-loop review steps.
- Evaluating LLM output quality, prompt design, and retrieval accuracy to reduce hallucination in live production use cases.
- Collaborating cross-functionally with data engineering and business teams to integrate AI capabilities into existing enterprise systems.

AI Internship · Fraunhofer IPK

Apr 2026 – Jun 2026 · Berlin, Germany

- Developed **Generative AI** applications using Python and **LLMs** for industrial automation.
- Designed and implemented **RAG pipelines** integrating structured engineering knowledge.
- Worked with vector databases, semantic search, embeddings, and context retrieval to improve response quality.
- Conducted prompt engineering, experimentation, and iterative LLM evaluation.
- Built reusable AI components following modular, production-ready software design.

Software Developer · CodingPro

Jul 2024 – Apr 2025 · Bengaluru, India

- Developed scalable Python applications supporting Generative AI and machine learning workflows.
- Built production-ready backend services using **FastAPI** and REST APIs.
- Designed reusable software modules for AI automation and enterprise deployment.
- Implemented model evaluation pipelines, experimentation workflows, and performance benchmarking.

Software Developer, Project Trainee · Indian Space Research Organisation

Feb 2024 – May 2024 · Bengaluru, India

- Developed NLP and speech-processing pipelines using Python.
- Built data preprocessing and feature engineering pipelines; evaluated ML models and documented results.

Full Stack Intern · **Municipal Corporation of Delhi**Jun 2023 – Jul 2023 · Delhi, India

- Developed backend applications integrating NLP, IoT, and geospatial datasets; built scalable APIs for data processing and analytics.

EDUCATION

M.Sc. in Artificial Intelligence · **Brandenburgische Technische Universität**Oct 2025 – Present · Germany

Neural Signal Analysis · Image Processing & Computer Vision · Causal Data Science · Modeling of Perception and Action

B.Sc. in Computer Science · **M.S. Ramaiah University of Applied Sciences**2021 – 2024 · Bengaluru, India

GPA: 1.9 / 4.0 (equivalent grading scale)

TECHNICAL SKILLS

Generative AI	LLMs, RAG & RAG Pipelines, LangChain, LlamaIndex, OpenAI API, Prompt Engineering, AI Agents & Agent Orchestration, MCP
Automation	n8n, Agentic Workflows, FastAPI, REST APIs
Vector Search	FAISS, ChromaDB, Vector Databases, Semantic Search, Embedding Models
Machine Learning	PyTorch, Scikit-learn, Pandas, NumPy, Explainable AI (SHAP, LIME)
Engineering	Python, SQL, Git, Docker, Linux, Testing, Agile, Technical Documentation

PROJECTS

Enterprise Knowledge Assistant **Python · LangChain · LlamaIndex · FastAPI · FAISS · ChromaDB**

- End-to-end RAG application with semantic search, document chunking, embedding generation, and metadata filtering.
- Modular FastAPI services with prompt-engineered OpenAI LLM inference for scalable enterprise deployment.

Explainable AI for Network Intrusion Detection **Python · PyTorch · Scikit-learn · SHAP · LIME**

- Deep neural network, Random Forest, and LightGBM models evaluated on CIC-IDS2017, NSL-KDD, and SIMARGL datasets.
- SHAP/LIME explanations for interpretability, benchmarked on accuracy, precision, recall, F1, and ROC-AUC.

AI Driver Drowsiness Detection **Python · OpenCV · PyTorch**

- Real-time computer vision system with optimized inference pipelines, evaluated across multiple testing scenarios.

PUBLICATIONS

Early Detection of Dyslexia Using Multimodal Analysis — INCOFT	02.2025
WeSAFE: Enhancing Human Security Mobile Application — IJIRCCE	12.2023

LANGUAGES & CERTIFICATES

Languages	Certificates
- English — C1 (Fluent) - German — A2–B1 (currently pursuing)	- Machine Learning Specialization — Stanford Online / DeepLearning.AI (3-course specialization) - Generative AI — Databricks